* Module 03 -- Hirely Backend Architecture Deep Dive

CONFIDENTIAL & PRIVATE STUDY MATERIAL
Target Roles: AI Engineer, Applied AI Engineer, Backend AI Engineer
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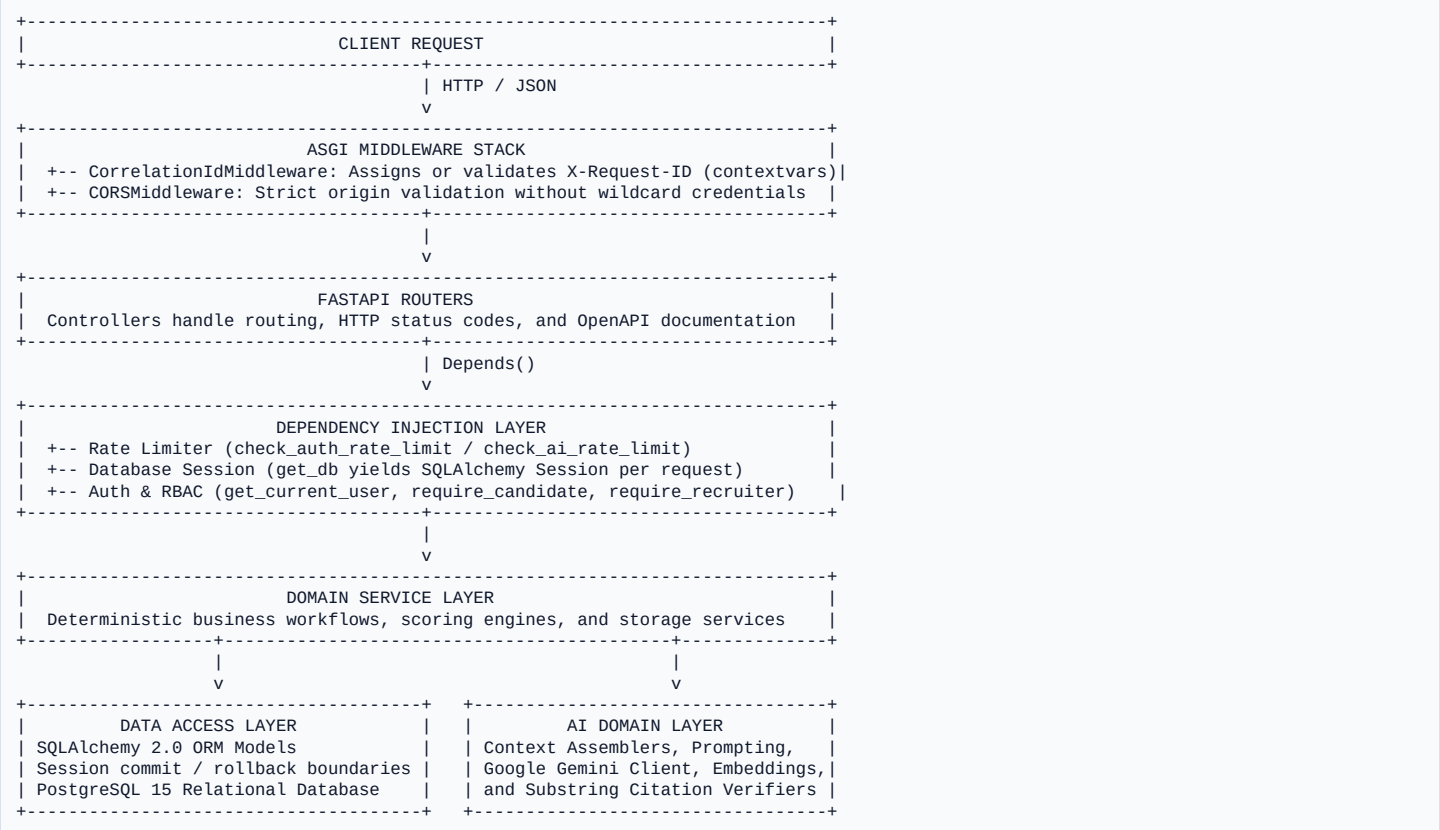
1. Backend Overview & Architecture Paradigm

The Hirely backend is built with **Python 3.12** and **FastAPI**, designed around a **layered, domain-driven architecture**. It deliberately separates:

1. **API Controllers (`app/api/`):** Route handling, HTTP status codes, and dependency injection.
2. **Business Services (`app/services/` & `app/ai/`):** Domain logic, orchestration, and model invocation.
3. **Data Access & Models (`app/models/` & `app/db/`):** SQLAlchemy ORM entities and database session lifecycles.
4. **Validation Contracts (`app/schemas/`):** Strict Pydantic v2 request/response schemas.
5. **Core Infrastructure (`app/core/` & `app/storage/`):** Configuration, rate limiting, security, observability, and storage providers.



2. Module Organization & Codebase Tree

```
backend/
+-- alembic/                        # Database migration environment
|   +-- versions/                   # 7 sequential schema migration scripts
|   +-- env.py                      # Alembic execution environment importing Base & settings
+-- app/
|   +-- ai/                         # AI domain submodules
|   |   +-- ats_analyzer/           # ATS scoring service, prompt templates, schemas
|   |   +-- candidate_intelligence/  # Candidate semantic filtering and scoring
|   |   +-- career_coach/           # Career coaching service and prompts
|   |   +-- cover_letter/           # Grounded cover letter generation
|   |   +-- embeddings/             # Gemini embedding service & cosine similarity
|   |   +-- eval/                   # Deterministic offline evaluation runner & datasets
|   |   +-- job_description_analyzer/# JD quality and structural clarity analyzer
|   |   +-- loaders/                # Text & document loaders
|   |   +-- matching/               # Two-sided applicant matching orchestrator
|   |   +-- parsers/                # Docling C++ document parser service
|   |   +-- prompts/                # Core prompt engineering & system instructions
|   |   +-- rag/                    # Candidate Resume Evidence Assistant RAG pipeline
|   |   +-- representations/        # Structured candidate & job text representations
|   |   +-- resume_improvement/     # Bullet rewrite & missing skill diagnosis
|   |   +-- services/               # Gemini client abstraction, token & retry logic
|   +-- api/                        # FastAPI route definitions
|   |   +-- applications.py         # /applications/me, /applications/recruiter, /status
|   |   +-- auth.py                 # /auth/register, /auth/login, /auth/me
|   |   +-- candidate.py            # /candidate/profile and candidate AI routes
|   |   +-- candidate_skills.py     # /candidate/skills CRUD
|   |   +-- dependencies.py         # FastAPI dependencies: get_db, auth, RBAC
|   |   +-- health.py               # /health and /health/db
|   |   +-- jobs.py                 # /jobs CRUD, /apply, /match
|   |   +-- job_skills.py           # /jobs/{job_id}/skills CRUD
|   |   +-- recruiter.py            # /recruiter/profile, /jobs, /description-analysis, RAG
|   |   +-- resumes.py              # /resumes upload, parse, activate, delete
|   +-- core/                       # Cross-cutting platform concerns
|   |   +-- observability/          # Contextvars, structured logging, latency metrics
|   |   +-- config.py               # Pydantic BaseSettings loading from .env
|   |   +-- rate_limit.py           # In-memory sliding-window rate limiters
|   |   +-- security.py             # Argon2 password hasher & JWT creation/verification
|   +-- db/                         # Database connection and session factory
|   |   +-- base.py                 # SQLAlchemy DeclarativeBase
|   |   +-- session.py              # Engine configuration, connection pooling, get_db
|   +-- models/                     # 10 SQLAlchemy ORM Entities
|   |   +-- user.py, candidate.py, recruiter.py, company.py, job.py,
|   |   +-- application.py, skill.py, candidate_skill.py, job_skill.py, resume.py
|   |   +-- __init__.py             # Central entity exports for Alembic
|   +-- schemas/                    # Pydantic validation contracts
|   |   +-- application.py, auth.py, candidate.py, job.py,
|   |   +-- recruiter.py, resume.py, skill.py
|   +-- services/                   # Core business logic services
|   |   +-- application.py, auth.py, candidate.py, candidate_skill.py,
|   |   +-- job.py, job_skill.py, recruiter.py, resume.py, resume_skill.py, skill.py
|   +-- storage/                    # File storage abstractions
|   |   +-- file_validation.py      # Multi-layer PDF security checks
|   |   +-- storage_service.py      # Local filesystem vs Cloudflare R2 provider
|   +-- main.py                     # FastAPI application factory & router registration
+-- tests/                          # Complete test suite (50+ unit/integration test files)
+-- alembic.ini                     # Alembic root config referencing backend/alembic
+-- Dockerfile                      # Python 3.12-slim production container image
+-- pytest.ini                      # Test configuration with pythonpath and custom markers
```

3. FastAPI Application Initialization & Middleware Chain

Hirely initializes the application in **`backend/app/main.py`**:

```
# 1. Initialize structured privacy-preserving observability
setup_observability(level=logging.INFO)

app = FastAPI(
    title=settings.app_name,
    description="Backend API for Hirely",
    version=settings.app_version,
    debug=settings.debug,
)

# 2. Correlation ID Middleware (Outer-most layer)
app.add_middleware(CorrelationIdMiddleware)

# 3. Dynamic CORS Middleware
cors_origins = [origin for origin in settings.cors_origins if origin]
has_wildcard = "*" in cors_origins
allow_credentials = False if has_wildcard else True

app.add_middleware(
    CORSMiddleware,
    allow_origins=cors_origins,
    allow_credentials=allow_credentials,
    allow_methods=["*"],
    allow_headers=["*"],
)
```

Critical Middleware Details:

1. **`CorrelationIdMiddleware`:** Intercepts incoming requests. Reads the **`X-Request-ID`** header, sanitizes it (stripping **`\r\n`** to prevent header injection), and injects it into thread-safe Python **`contextvars`**. If missing or invalid, it mints a fresh UUID4.
2. **`CORSMiddleware` Security Guard:** Dynamic origins are loaded from **`CORS_ORIGINS`**. Crucially, the code enforces: **Never combine credentialed requests (`allow_credentials=True`) with a wildcard (`*`) origin**, preventing cross-origin session hijacking.

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4. API Request Lifecycle & Layered Boundaries

To see how the boundaries work in practice, let's trace the execution flow when a candidate applies to a job:

```
sequenceDiagram
    autonumber
    actor Candidate as Client (Browser)
    participant ASGI as Middleware Chain
    participant Router as jobs.py (Router)
    participant Dep as dependencies.py (RBAC)
    participant Service as application.py (Service)
    participant DB as PostgreSQL (via SQLAlchemy)

    Candidate->>ASGI: POST /jobs/{job_id}/apply (Header: Bearer JWT)
    ASGI->>ASGI: CorrelationIdMiddleware sets X-Request-ID
    ASGI->>Router: Route matched to apply_to_job(job_id)
    Router->>Dep: Depends(require_candidate)
    Dep->>Dep: Decode JWT, verify role == "candidate", check user.is_active
    Dep-->>Router: Current User injected
    Router->>DB: Query Candidate by user_id
    Router->>DB: Query Job by job_id (is_active == True)
    Router->>Service: create_application(db, candidate, job)
    Service->>DB: SELECT Application WHERE candidate_id AND job_id
    alt Application Already Exists
        Service-->>Router: Raise ValueError("Already applied")
        Router-->>Candidate: HTTP 409 Conflict
    else Application Valid
        Service->>DB: INSERT INTO applications (status='applied')
        Service->>DB: db.commit() & db.refresh()
        Service-->>Router: Return Application ORM instance
        Router-->>Candidate: HTTP 201 Created (ApplicationResponse JSON)
    end
```

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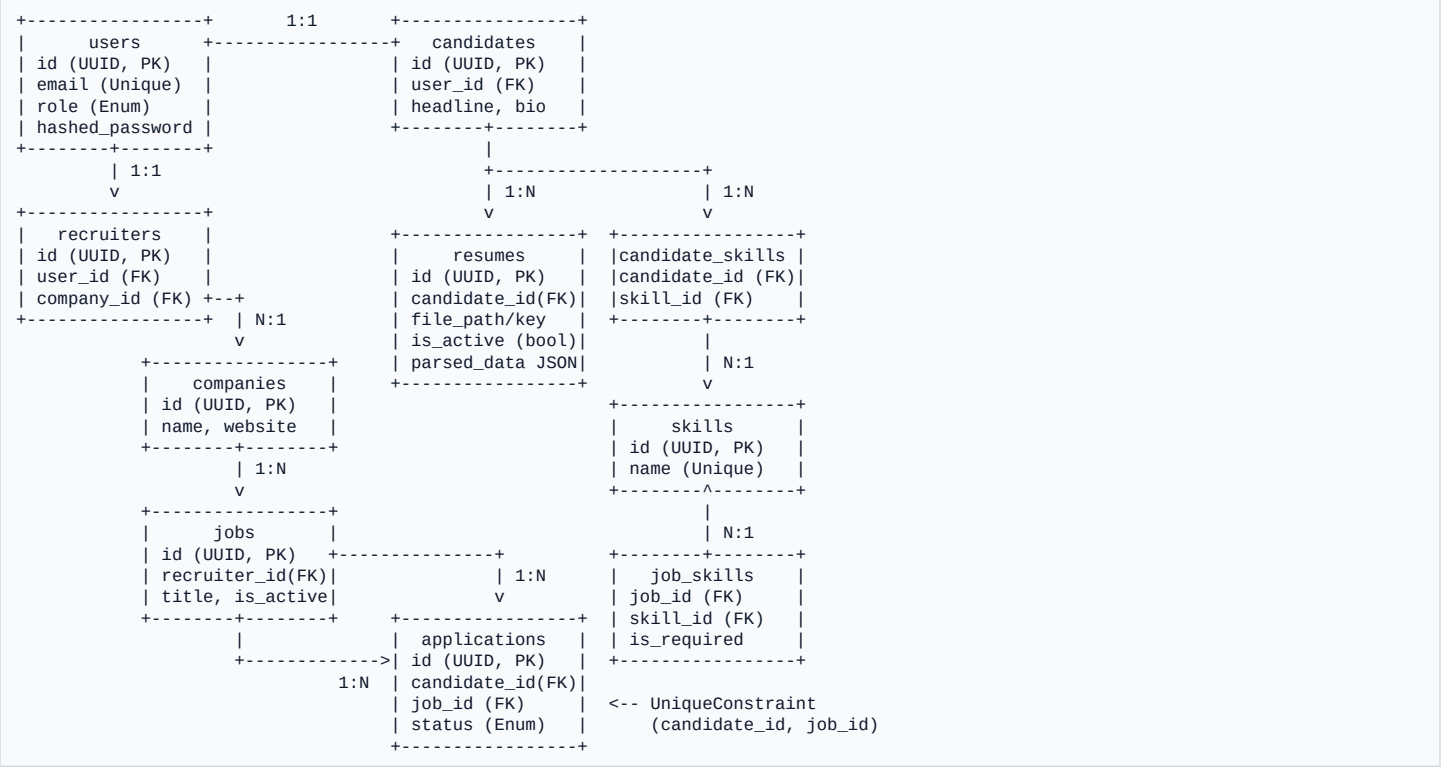
5. PostgreSQL Database Schema & Relational Integrity

The database comprises **10 relational tables** defined in **`backend/app/models/`**:



Critical Database Integrity Rules:

1. **Duplicate Application Prevention:** `UniqueConstraint("candidate_id", "job_id", name="uq_candidate_job")` guarantees an applicant can never submit multiple applications to the same opening.
2. **Foreign Key Cascades:** Deleting a candidate automatically cleans up their `candidate_skills` and `resumes` records.
3. **Application Status State Machine:** Enforced via `ApplicationStatus` Enum:

$$applied \longrightarrow shortlisted \longrightarrow interview \longrightarrow hired \text{ or } rejected$$

Terminal states are `hired` and `rejected`.

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6. SQLAlchemy 2.0 ORM & Session Lifecycle Management

Database connection pooling and session lifecycle are managed in **`backend/app/db/session.py`**:

A. Engine Configuration & Connection Pool Budgeting

```
engine = create_engine(
    settings.database_url,
    pool_size=settings.db_pool_size,          # Default: 5
    max_overflow=settings.db_max_overflow,    # Default: 10
    pool_timeout=settings.db_pool_timeout,    # Default: 30s
    pool_recycle=settings.db_pool_recycle,    # Default: 1800s (30m)
    pool_pre_ping=settings.db_pool_pre_ping,  # Default: True (tests connection liveness)
)
```

Worker Connection Formula:

$$Max\ PostgreSQL\ Connections = (DB_POOL_SIZE + DB_MAX_OVERFLOW) \times Uvicorn\ Workers$$

$$Max\ Connections = (5 + 10) \times 4 = 60$$

This guarantees that 4 workers consume at most 60 slots out of PostgreSQL's default `max_connections = 100`, reserving 40 connections for schema migrations, superuser administration (`superuser_reserved_connections = 3`), and monitoring.

B. Session Lifecycle Pattern (`get_db`)

```
def get_db():
    db = SessionLocal()
    try:
        yield db
    finally:
        db.close()
```

FastAPI automatically yields the database session to the router dependency. When the HTTP request finishes (whether with success or an unhandled exception), the `finally` block executes, returning the connection back to the SQLAlchemy pool and preventing connection leaks.

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7. Alembic Migration Strategy & Version History

Hirely maintains 7 sequential, version-controlled schema migrations under `backend/alembic/versions/`:

Revision ID	Description	Created Tables / Schema Changes
`686f5e817bad`	Create initial user profiles	`users`, `candidates`, `recruiters`
`86847c4ad53a`	Add companies & recruiter relation	`companies` table, `recruiter.company_id` foreign key
`2a3d08cd70c2`	Clean recruiter company name	Drops redundant `company_name` string column from recruiters
`8b276abe3679`	Add jobs table	`jobs` table with title, description, location, recruiter foreign key
`1fcd1e35fb2f`	Add applications table	`applications` table with state machine enum and unique constraint
`0001d605bd8c`	Add skills tables	`skills`, `candidate_skills`, and `job_skills` association tables
`ed9389445137`	Add resumes table	`resumes` table with storage paths, parsing status, and JSON data

In Docker Compose, the `hirely-migration` container runs `alembic upgrade head` automatically before the backend starts, ensuring zero migration race conditions.

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8. Authentication, Argon2, JWT & Role-Based Access Control

A. Password Hashing with Argon2

Hirely uses `argon2-cffi` inside `backend/app/core/security.py`:

- **Why Argon2 instead of bcrypt?** Argon2 was the winner of the Password Hashing Competition (PHC). It is memory-hard, making it resistant to GPU/ASIC hardware brute-force attacks.
- Passwords are encrypted before database insertion: `ph.hash(password)`.
- Passwords are never stored in plaintext or logged.

B. Stateless JWT Bearer Authentication

- **Token Generation:** Encodes `sub` (User ID as string), `role` (`candidate` or `recruiter`), and `exp` (UTC timestamp + 30 minutes).
- Signed using `HS256` with a minimum 32-character `JWT_SECRET_KEY`.

C. Role-Based Access Control (RBAC) Dependencies

FastAPI dependencies in `backend/app/api/dependencies.py` enforce role boundaries:

```
def require_candidate(current_user: User = Depends(get_current_user)) -> User:
    if current_user.role != UserRole.CANDIDATE:
        raise HTTPException(
            status_code=status.HTTP_403_FORBIDDEN,
            detail="Access forbidden: candidates only",
        )
    return current_user
```

If a recruiter attempts to call `/candidate/profile` or `/candidate/ats-analysis`, FastAPI rejects the request with **HTTP 403 Forbidden** before any controller or service code executes.

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9. Resume Upload, Multi-Layer Validation & Storage Engine

Resume uploads follow a strict security and processing pipeline in `backend/app/api/resumes.py`:

```
Candidate Upload (PDF File)
        |
        v
[File Validation: validate_resume_file()]
  +-- Magic Byte Check: First 5 bytes MUST match b"%PDF-" (rejects renamed .exe or .html)
  +-- Content-Type Check: Must be application/pdf
  +-- Max Upload Size Check: Stream bounded to 10 MB
  +-- ZIP Bomb Protection: Max 1,000 ZIP entries, max 50 MB uncompressed
  +-- Page Limit: Docling configured with max_num_pages = 10
        |
        v
[Storage Service: storage_service.save_resume()]
  +-- If STORAGE_BACKEND == "local":
  |   Writes file to local filesystem volume (e.g. app/storage/resumes/<uuid>.pdf)
  +-- If STORAGE_BACKEND == "r2":
      Streams file to Cloudflare R2 bucket via boto3 (S3 API key: resumes/<uuid>.pdf)
        |
        v
[Database Record: create_resume()]
  Inserts row into resumes table: file_path, parsing_status="pending", is_active=False
        |
        v
[Parse Resume: parse_resume_endpoint()]
  Calls Docling ResumeParserService:
  +-- Extracts raw text and detects layout sections
  +-- Identifies skills, experience bullets, education
  +-- Saves structured JSON into resume.parsed_data
  +-- Syncs discovered skills to candidate_skills table
  +-- Invalidates in-memory candidate embedding cache
```

10. Error Handling Strategy & Exception Classification

To ensure production safety without leaking internal stack traces or database connection strings to end users:

1. **Standard HTTP Exceptions:** Expected business rejections use explicit `HTTPException` codes:

- `400 Bad Request`: Validation failure, empty job description, inactive job.
- `401 Unauthorized`: Invalid password, expired JWT token.
- `403 Forbidden`: Role mismatch (recruiter accessing candidate endpoint).
- `404 Not Found`: Entity (Job, Candidate, Resume) does not exist.
- `409 Conflict`: Duplicate application, email already registered.

2. **Exception Classification (`classify_exception`):**

In AI endpoints, caught exceptions are categorized into fixed enums (`AIErrorCategory.TIMEOUT`, `RATE_LIMIT`, `MODEL_UNAVAILABLE`, `UPSTREAM_SERVER_ERROR`).

3. **Structured Error Logging:**

```
   category, exc_name = classify_exception(exc)
   logger.error(
       "ATS Analysis service failure: %s",
       exc_name,
       extra={
           "correlation_id": get_correlation_id(),
           "feature": AIFeature.ATS_ANALYZER.value,
           "error_category": category.value,
           "exception_type": exc_name,
       },
   )
   raise HTTPException(
       status_code=status.HTTP_502_BAD_GATEWAY,
       detail="ATS Analysis service is temporarily unavailable.",
   ) from exc
```

The client receives a clean **HTTP 502 Bad Gateway** message, while the monitoring system logs a sanitized, structured JSON event with the exact correlation ID for debugging.

11. Architectural Decisions & Alternatives Analysis

Architectural Decision	Chosen Strategy	Alternative Rejected	Rationale
Pydantic v2 over Marshmallow	Pydantic v2 (compiled in Rust)	Marshmallow / Cerberus	Pydantic v2 is natively integrated into FastAPI, provides automatic OpenAPI documentation, and executes type conversions ~5x faster.
SQLAlchemy 2.0 2.0-Style Queries	`db.execute(select(Job).where(...))` / Session ORM	Raw SQL cursor queries	Raw SQL lacks type-safe entity mapping, requires manual object reconstitution, and is vulnerable to injection if developer sanitization slips.
Synchronous DB Sessions over Async	`SessionLocal()` (Synchronous Psycopg 3)	`AsyncSession` (AsyncPG)	In CPU-bound AI preprocessing and relational lookups with connection pooling, synchronous SQLAlchemy sessions are battle-tested, simpler to debug, and avoid complex async greenlet thread-pooling bugs in Alembic.
Dual Storage Adapter Pattern	Abstract `StorageService` (`local` vs `r2`)	Hardcoded local disk paths	Hardcoding disk paths breaks containerized cloud hosting (Render/Fly/Heroku) where container filesystems are ephemeral. The adapter allows switching between local dev and cloud object storage via a single `.env` flag.